

Lab 3 – Component Modelling & Architectural Pattern Selection

Architecture Selection & Justification

Automated Rubric Assignment Evaluator

Architecture Selection

We chose Microservices Architecture for the Automated Rubric Assignment Evaluator System.

Architectural Choice

The selected architectural style is Microservices Architecture, where the system is divided into small, independently deployable services. This is suitable for the evaluator because submission handling, queue management, code execution, rubric scoring, and result/review functions can operate as separate services.

Two Reasons

- 1. Independent scaling:** The system must handle up to 100 concurrent project submissions. Microservices allow high-load services such as the submission queue and code execution/test runner to be scaled independently instead of scaling the entire application.
- 2. Fault isolation and independent deployment:** A failure in one evaluation service can be isolated from other services, while individual services can be updated or deployed independently. This supports reliable batch processing and reduces the impact of failures on the complete evaluation workflow.

Security Advantage

Microservices allow security controls to be applied separately to sensitive services. In this system, the Code Execution & Test Runner can be isolated and run student submissions in a controlled sandbox, limiting the impact of untrusted project code on the rest of the evaluator.

Performance Benefit

Microservices provide independent scaling and help distribute evaluation work across services. Combined with the submission queue, this supports concurrent batch processing and helps the system meet the target of returning the rubric breakdown with test results within 10 seconds of submission.